SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

---

COURSE OUTCOME COVERED

CO2: Apply suitable Data Pre processing techniques like Data cleaning, Data intergration,Data transformation and data Reduction , for effective data preparation (BL3,BL4)
Assessment Tool Weightage: 50%

QUESTIONS: 5 **Total Marks:50**

Annexure B
Analytical Problem Solving

Criteria	Problem Understanding ***(2)***	Method Selection ***(2)***	Solution Process ***(2.5)***	Accuracy of Calculation ***(2)***	Interpretation of Results ***(1.5)***	***Total(10)***
Weightage	*20%*	20%	25%	20%	15%	*100%*
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I _V.Dharanireddy(192311219)___________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: v.dharanireddy

RUBRICS FOR CALCULATION:

Numerical Problems					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Annexure B

Analytical Problem Solving

1. Covariance Analysis in Health Insurance (10 Marks)

The table below shows the **annual premium increase rate (xi)** and the **number of claims per 100 policyholders (yi)** in a health insurance company.
Using the covariance formula, determine whether premium increase and claim frequency have a **positive or inverse relationship**.
Before computing covariance, calculate the **mean of x and y**.

Premium Increase % (xi) Claims per 100 Policyholders (yi)

3.2 15 4.0 18 5.5 25

4.8 20

31/07/26
Friday

Assessment - 01 - Chapter - 02

CSA1613 - (DWDM)

V. Dharani reddy
192211219
(CSE)

1(a) Covariance Analysis in Health insurance, The table - per 100 policyholders in a health insurance company - using covariance formula, determine whether premium increase and claim frequency have a positive inverse relationship. premium increase (%) (xi) claims per 100 (x°)?

Answer:

Given data:

Premium Increase (%)(xi):	Claims per 100 policyholders:
3.2	15
4.0	18
5.5	25
4.8	20

Step 1: Calculate the Mean

$\bar{x} = \frac{3.2+4.0+5.5+4.8}{4} = \left(\frac{17.5}{4}\right) = (4.375)$

$\bar{y} = \frac{15+18+25+20}{4} = \left(\frac{78}{4}\right) = (19.5)$

Coviance Table - (Step 2)

x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
3.2	15	-1.175	-4.5	5.2875
4.0	18	-0.375	-1.5	0.5625
5.5	25	1.125	5.5	6.1875
4.8	20	0.425	0.5	0.2125.

$\Sigma(x-\bar{x})(y-\bar{y}) = (5.2875 + 0.5625 + 6.1875 + 0.2125)$

$= (12.25)$

Step 3: Calculate Covariance;

using population Covariance formula:

$$Cov(x,y) = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{n} = \left(\frac{12.25}{4}\right) = (3.0625)$$

Final Answer:

mean of premium increase = $(\bar{x}) = (4.375\%)$

Mean of claims $(\bar{y}) = (19.5)$

Covariance (=) (3.0625)

2(c) Data smoothing using Equal-frequency Binning (HealthCare).
A hospital is analyzing patient blood pressure readings to identify trends due to minor fluctuations, the analyst performs local smoothing using equal-frequency binning?

Answer:

Given data;

110, 112 . . . . . ;

There are 27 values, divide into 3 equal-frequency bins (9 values each)

Bin 1:

110, 112, 115, 115, 118, 220, 120, 122, 124.

Bin 2:

124, 126, 126, 126, 126, 130, 135, 135, 138

Bin 3:

138, 138, 138, 140, 145, 120, 150, 152, 160, 180.

Smoothing by Bin Means:

2. Data Smoothing using Equal-Frequency Binning (Healthcare) (5 Marks)

A hospital is analyzing **patient blood pressure readings (systolic)** to identify trends. Due to minor fluctuations, the analyst performs **local smoothing using equal-frequency (equi-depth) binning**.

Sample Data Set (Systolic BP readings):

110, 112, 115, 115, 118, 120, 120, 122, 124, 124, 126, 126, 126, 126, 130, 135, 135, 138, 138, 138, 138, 140, 145, 150, 152, 160, 180

$\Sigma(x-\bar{x})(y-\bar{y}) = (5.2875 + 0.5625 + 6.1875 + 0.2125)$

$= (12.25)$

Step 3: Calculate Covariance;

using population Covariance formula:

$Cov(x,y) = \frac{\Sigma(x-\bar{x})(y-\bar{y})}{n} = \left(\frac{12.25}{4}\right) = (3.0625)$

Final Answer:

mean of premium increase $= (\bar{x}) = (4.375\%)$

Mean of claims $(\bar{y}) = (19.5)$

Covariance (=) (3.0625)

2(e) Data Smoothing using Equal-frequency Binning (Healthcare).
A hospital is analyzing patient blood pressure readings to identify trends due to minor fluctuations, the analyst performs local smoothing using equal-frequency binning?

Answer:

Given data;

110, 112 . . . . . ;

There are 27 values, divide into 3 equal-frequency bins (9 values each)

Bin 1:

110, 112, 115, 115, 118, 120, 120, 122, 124.

Bin 2:

124, 126, 126, 126, 126, 130, 135, 135, 138

Bin 3:

138, 138, 138, 140, 145, 120, 150, 152, 160, 180.

Smoothing by Bin Means:

Bin1 mean = (110+112+115+115+118+120+120+122+124 /9) = $\boxed{117.33}$

Bin2 mean = (124+126+126+126+126+130+135+135+138/9) = $\boxed{129.56}$

Bin3 mean = (138+138+138+140+145+150+152+160+180/2) = $\boxed{149.00}$

Smoothed Data:

117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33 → 1.

129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56 → 2

149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00 → 3

---

34. Equal width Binning for Risk Categorization.

An insurance company evaluate blood mass index of 18 clients for risk classification. To simplify analysis, they smooth data by replacing values with bin boundaries.

Answer:

Given BMI Data:

18.5, 19.2, 21.8 . . . . →

(minimum = 18.5), (maximum = 39.5)

(Range = 21)

Using 3 equal-width bins:

Bin 1 : 18.5 – 25.5

Bin 2 : 25.5 – 32.5

Bin 3 : 32.5 – 39.5

smoothing by closet: (Bin Boundary).

Bin 1: (18.5, 18.5, 25.5, 25.5)

3. Equal-Width Binning for Risk Categorization (5 Marks)

An insurance company evaluates **Body Mass Index (BMI)** of 18 clients for risk classification.
To simplify analysis, they smooth the data by replacing values with **closest bin boundaries using equal-width partitioning**. **Sample Data Set (Sorted BMI):**
18.5, 19.2, 21.8, 24.9, 25.5, 26.2, 26.4, 27.8, 29.2, 30.2, 30.4, 31.9, 32.4, 33.1, 33.6, 34.7, 38.2, 39.5

Bin1 mean = $(110+112+115+115+118+120+120+122+124/9) = \boxed{117.33}$

Bin2 mean = $(124+126+126+126+126+130+135+135+138/9) = \boxed{129.56}$

Bin3 mean = $(138+138+138+140+145+150+152+160+180/2) = \boxed{149.00}$

Smoothed Data:

117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33, 117.33 → 1

129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56, 129.56 → 2

149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00, 149.00 → 3

3Q. Equal width Binning for Risk Categorization.

An insurance Company evaluate blood mass index of 18 clients for risk classification. To simplify analysis, they smooth data by replacing values with bin boundaries.

Answer:

Given BMI Data:

18.5, 19.2, 21.8 . . . . ;

(minimum = 18.5), (maximum = 39.5)

(Range = 21)

Using 3 equal-width bins:

Bin 1 : 18.5 – 25.5

Bin 2 : 25.5 – 32.5

Bin 3 : 32.5 – 39.5

Smoothing by closet : (Bin Boundary).

Bin 1 : (18.5, 18.5, 25.5, 25.5)

Bin2: 25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5
Bin3: 32.5, 32.5, 32.5, 39.5, 39.5.

Final smoothed Data:

18.5, 18.5, 25.5, 25.5, 25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5
32.5, 32.5, 32.5, 32.5, 32.5, 32.5, 39.5, 39.5

— x —

4(f) Data reduction using Bin medians (median supply costs)
A hospital administration is reviewing cost of medical consumables To reduce for mining purpose, they discretize the sorted records using bin medians.

Answers:

First sorted data:

120, 140, 145, 145, 148, 150 . . . . . 300;

Divide into 3 equal parts (frequency bins)

Bin1:

120, 140, 145, 145, 148, 150, 150, 155.

median = (148)

Bin2:

155, 155, 155, 160, 160, 165, 170, 170, 175

median = (160)

Bin3:

4. Data Reduction using Bin Medians (Medical Supply Costs) (5 Marks)

A hospital administration is reviewing the **cost of medical consumables (in ₹)**. To reduce data for mining purposes, they discretize the sorted price records using **bin medians**.

Sample Data Set (Costs in ₹):

120, 150, 160, 160, 140, 145, 145, 148, 150, 150, 155, 155, 155, 155, 165, 170, 170, 175, 175, 175, 175, 180, 190, 200, 210, 230, 300

Bin2: 25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5, 32.5
Bin3: 32.5, 32.5, 32.5, 39.5, 39.5.

Final smoothed Data:

18.5, 18.5, 25.5, 25.5, 25.5, 25.5, 25.5, 25.5, 32.5, 32.5, 32.5, 32.5
32.5, 32.5, 32.5, 32.5, 32.5, 32.5, 39.5, 39.5

= x =

4(e) Data reduction using Bin medians (median supply costs)
A hospital administration is reviewing cost of medical consumables To reduce for mining purpose, they discretize the sorted records using bin medians.

Answers:

First sorted data:

120, 140, 145, 145, 148, 150 . . . . . 300;

Divide into 3 equal parts (frequency bins)

Bin1:

120, 140, 145, 145, 148, 150, 150, 155.

median = (148)

Bin2:

155, 155, 155, 160, 160, 165, 170, 170, 175

median = (160)

Bin3:

175, 175, 175, 180, 190, 200, 210, 230, 300

median = (190)

Data after Bm median smoothing;

148, 148, 148, 148, 148, 148, 148, 148, 148 → (Bm 1)

(Bm 2) → 160, 160, 160, 160, 160, 160, 160, 160, 160,

(Bm3) → 190, 190, 190, 190, 190, 190, 190, 190, 190.

— x —

5(a) Strandarization of patient Age Data;

Given the following ages of insured individuals;

Standerize variable by performing following:

a) Compute the ~~man~~ mean St abosolute deviation (MAD) of age.

b) Compute the Z-source the first four observation.

Solution:

Data:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

a) mean absolute Deviation (MAD)

---

5. Standardization of Patient Age Data (5 Marks)

Given the following **ages of insured individuals**:
18, 22, 25, 42, 28, 43, 33, 35, 56, 28

Standardize the variable by performing the following:
(a) Compute the **mean absolute deviation (MAD)** of age.
(b) Compute the **z-score for the first four observations**.

175, 175, 175, 180, 190, 200, 210, 230, 300

median =(190)

Data after Bm median smoothing;

148, 148, 148, 148, 148, 148, 148, 148, 148 → (Bm 1)

(Bm 2) → 160, 160, 160, 160, 160, 160, 160, 160, 160,

(Bm3) → 190, 190, 190, 190, 190, 190, 190, 190, 190.

— X —

5(e) Strandarization of patient Age Data;

Given the following ages of insured individuals;

Standarize variable by performing following:

a) Compute the ~~man~~ mean St abosolute deviation (MAD) of age.

b) Compute the Z-source the first four obserration.

Solution:

Data:

18, 22, 25, 42, 28, 43, 33, 35, 56, 28

a) mean absolute Deviation (MAD)

a) Mean:

$\bar{x} = \left(\frac{18+22+25+42+28+43+33+35+56+28}{10}\right)$

$= (33)$

$\therefore (\bar{x} = 33)$

Absolute deviations:

Age:	(Age − 33)
18	15
22	11
25	8
42	9
28	5
43	10
33	0
35	2
56	23
28	5

$\therefore$ sum $= (88)$

$(MAD) = \left(\frac{88}{10}\right) = (8.8)$

b) z-scores for first four observations:

SD : $(\sigma) = \sqrt{\frac{1014}{10}} = \sqrt{101.4} = (10.07)$

$z = \frac{x - \bar{x}}{\sigma}$

Age:		z-score:
18	−	−1.49
22	−	−1.09
25	−	−0.79
42	−	0.89

Final Answers:

Mean = 33

MAD = 8.8

z score ÷ (18) → (−1.49)
(22) → (−1.09)
(25) → (−0.79)
(42) → (0.89)